OpenADMET CYP Challenge — Structure Prediction Track Model Description Report

Submitted by: Cheminfosolutions Ltd · **leaderboard alias:** Cheminfo
Submission: 20 CYP3A4 protein–ligand complexes predicted from ligand SMILES alone; one archive of flat `{Molecule_Name}.pdb` files, chain A protein, chain B haem, chain C ligand as residue `LIG`
Metrics: LDDT-PLI (primary), BiSyRMSD (secondary), LDDT-LP (protein), with PoseBusters as a validity gate

1. Methods

No receptor structure is provided, so each complex is produced from sequence and SMILES. Two pose sources were built — co-folding and ensemble docking into the co-folded receptors — and the submitted archives draw from both.

Component	Method / configuration	Role	Reference
Boltz-2 co-folding	v2.2.1; UniProt P08684 canonical sequence, 503 residues, sequence and MSA only, no structural template; haem B as CCD `HEM` so atom names match the references; 24 soft distal pocket contacts at 8 Å, Cys442 excluded as proximal-face; steering potentials on; 5 diffusion samples × 3 seeds × 20 ligands = 300 complexes	Protein fold; non-chelator poses; receptor ensemble for docking	Boltz-2 (Passaro et al., 2025 preprint); MIT
MSA	14,302 sequences, computed once and shared across every ligand and seed	Input to co-folding	ColabFold MSA server (Mirdita et al., *Nat. Methods* **2022**, 19, 679–682)
gnina ensemble docking	v1.3.3; the 300 co-folded complexes stripped to protein + haem as a receptor ensemble; identical 24 Å box in every frame, centred 7.0 Å along the haem distal normal; `--cnn_scoring rescore --num_modes 9 --exhaustiveness 8 --seed 42`; 6,000 runs → 50,509 poses	Submitted poses for the coordinating ligands	McNutt et al., *J. Cheminform.* **2021**, 13, 43
Covalent iron pin	12 ligands with a coordinating nitrogen pinned to an axial site 2.10 Å along the distal haem normal, by per-ligand smallest rooted-environment SMARTS; 28,909 pinned poses retained, re-measured from written coordinates at 2.100 Å	Haem-iron coordination	gnina covalent mode (as above)
Pose selection	Rank-1 CNNaffinity with picks forced into distinct protein conformations (submitted archives). Also implemented: Boltz `ligand_iptm` arg-max, and haem-superposed medoid — the pose of lowest mean RMSD to all others	One pose per ligand	internal
Validity repair	PoseBusters 0.6.4 in `dock` mode used as the oracle: walk the CNNaffinity-ranked list in distinct receptors, export 12 candidates, run the gate over the batch, take the highest-scoring candidate that passes; a pinned pose is eligible only if its pin fired at 2.10 ± 0.25 Å	Brings every archive to full validity	Buttenschoen, Morris & Deane, *Chem. Sci.* **2024**, 15, 3130–3139
Support	RDKit 2025.3.6 / 2025.9.6 (ligand handling, SMARTS, atom naming); gemmi 0.7.4 / 0.7.5 (CIF and PDB I/O); MDAnalysis 2.9.0 (challenge validator)	Archive construction and validation	respective projects

Hardware. Co-folding on an NVIDIA Tesla T4 (16 GB) at one worker; docking on a local RTX 4060 at 16 workers, ~25 runs per minute free and ~3× that cost per pinned run. A 503-residue complex with haem is ~4,100 atoms and needs 6–10 GB VRAM per worker.

2. Results

Archive	Date (UTC)	Method	PoseBusters	LDDT-PLI
seed1	2026-10-13	Co-folding, 1 seed, confidence pick	20 / 20	0.4106

Archive	Date (UTC)	Method	PoseBusters	LDDT-PLI
run2	2026-10-15	Co-folding, 3 constrained seeds, medoid pick	20 / 20	0.4137
dock	2026-10-26	All gnina rank-1	15 / 20	0.3387
clean20	2026-10-27	**gnina rank-1 + 5 repaired poses**	**20 / 20**	**0.4300**
hybridH	2026-10-28	Co-folded run2 + clean20's 5 repairs	20 / 20	0.4111
hybridHC	2026-10-28	clean20 chelators + co-folded non-chelators	20 / 20	0.4293
geomfix	2026-11-02	clean20 with 2 geometry re-picks	20 / 20	0.3926

Haem-iron coordination, measured over the 10 ligands carrying a coordinating nitrogen: co-folded poses 2.83–2.87 Å, pinned docked poses 2.100 Å, organisers' baseline 2.00–2.14 Å. Mean ligand–iron distance after validity repair 2.970 Å, from 2.895 Å before. Archive checksums are not reproducible — a rebuild gives all twenty PDB files byte-identical but a different archive hash, as the format embeds modification times — so archives are retained as artefacts.

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3. Ligand and receptor preparation

Ligand protonation	OpenBabel pH 7.4 model, SMARTS-audited per ionisable group; all twenty ligands neutral, confirmed two independent ways
Protonation tool choice	dimorphite-dl not used: at pH 7.4 it returns protonated amide, urea and sulfonamide nitrogens and a net +3 charge on one ligand
Coordinating-atom selection	Per-ligand smallest rooted-environment SMARTS. gnina pins the first atom of the first SMARTS match and cannot pin by index, so each pattern is rooted at the intended nitrogen
Receptor	UniProt P08684 canonical sequence, 503 residues, with a 14,302-sequence MSA and no structural template; haem B supplied as the CCD `HEM` entry
Docking receptors	Each of the 300 co-folded complexes stripped to protein plus haem, its own ligand discarded
Search box	Fixed identically in every frame rather than auto-boxed: a 24 Å cube centred 7.0 Å along the haem distal normal from the iron. Distal face identified in 300 of 300 frames; the box encloses each frame's native ligand in 300 of 300
Archive format	Flat `{Molecule_Name}.pdb` per ligand; chain A protein, chain B haem, chain C ligand as residue `LIG`; unique (chain, resid, icode) keys, atom names and atom serials throughout